

Observation Models in Evolutionary Predator–Prey Robotics: An Exploratory Comparison of Privileged, 360° Camera, and First-Person View Observations

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Abstract. Predator–prey robotics provides a useful setting for studying the evolution of pursuit and coordination in multi-agent systems. However, evolved controllers are often provided with exact simulator state information that would not normally be available to physical robots. This thesis compares the controller evolution and performance observed under three observation representations.

Three observation models are compared: privileged geometric information, 360° camera observations, and first-person view (FPV). Three simulated Thymio predators use a shared feedforward neural-network controller whose parameters are optimized using the Covariance Matrix Adaptation Evolution Strategy. The environment, prey controller, fitness function, network architecture, and evolutionary settings are held constant across the three conditions.

Controller fitness increased during all three evolutionary runs. In the final evaluation, the selected privileged controller achieved a mean fitness of 3.07 ± 0.52 , followed by the 360° camera controller with 2.89 ± 0.45 and the FPV controller with 2.65 ± 0.42 . However, a Kruskal–Wallis test did not detect a statistically significant difference between the final fitness-score distributions ($H(2) = 2.511$, $p = 0.285$).

The results suggest that direct geometric information and broader visual coverage may benefit evolved pursuit behaviour under the tested conditions. However, because only one evolutionary run was conducted for each observation model, the study provides an exploratory comparison rather than a general statistical ranking of the three representations.

Keywords: Evolutionary robotics · Predator–prey robotics · Observation models · CMA-ES · Multi-robot systems

1 Introduction

Evolutionary robotics investigates how robot controllers and behaviours can be generated through evolutionary optimization rather than designed entirely by hand. Predator–prey tasks are particularly useful in this field because they combine pursuit, evasion, adaptation, and interaction between multiple autonomous agents. When several predators pursue a faster or more manoeuvrable prey, successful behaviour may also require the predators to coordinate their movements and occupy complementary positions.

The behaviour that can evolve depends strongly on the information available to the controller. In simulation, controllers can be provided with privileged information such as the exact positions, distances, and relative directions of other robots. Such observations simplify perception and provide a clear description of the environment. However, this information would generally not be directly available to physical robots, which must instead rely on onboard sensors such as cameras. Camera-based observations are less precise and may provide only a partial view of the environment, but they represent a more realistic sensing assumption.

The difference between privileged and camera-based observations creates an important experimental question. A controller receiving exact geometric information may evolve effective pursuit behaviour more easily, while a controller using visual input must act on approximate features derived from camera images. The field of view may also influence performance. A panoramic camera representation can provide visual information from most directions, whereas a front-facing camera leaves the controller partially observable when the prey or neighbouring predators move outside the image.

As discussed in Section 2, the literature reviewed in this thesis covers controller evolution, cooperation, communication, competitive co-evolution, and sim-to-real transfer. Across these studies, the observation model is generally treated as part of the experimental setup rather than as the primary variable of comparison. Direct controlled comparisons between privileged information, panoramic vision, and restricted first-person vision therefore appear to be limited.

This thesis addresses this gap by comparing three observation models in the same simulated predator–prey task: privileged geometric observation, 360° camera observation, and first-person view (FPV). The experiment uses three simulated Thymio predators and one prey controlled by a fixed controller in a bounded arena. All predators share the same feedforward neural-network controller, and the network parameters are optimized using the Covariance Matrix Adaptation Evolution Strategy. The simulation environment, controller architecture, prey behaviour, fitness function, initial configuration, and evolutionary settings are kept constant so that the observation representation remains the primary difference between the experimental conditions.

The comparison focuses on whether effective pursuit-like behaviour can evolve under each observation model and on the performance of the selected controllers during repeated evaluation. Because only one evolutionary run is conducted for each condition, the study is exploratory. It compares the controllers produced by the three observed runs but does not aim to establish a general statistical ranking of the observation models.

1.1 Research Question

The thesis addresses the following research question:

RQ: What differences in evolved predator-controller performance are observed between privileged, 360° camera, and first-person view observation models in a simulated predator–prey task?

The study contributes a controlled implementation and comparison of three observation representations using an otherwise shared experimental setup. It also examines evolutionary training progress, repeated controller performance, and representative predator trajectories in order to provide both quantitative and qualitative insight into the resulting behaviours.

The remainder of this thesis is structured as follows. Section 2 reviews previous research on predator–prey robotics, multi-agent cooperation, and simulation-based evolutionary robotics. Section 3 describes the simulation, observation models, controllers, fitness function, and evaluation procedure. Section 4 presents the experimental results, followed by their interpretation and limitations in Section 5. Section 6 concludes the thesis.

2 Related Work

2.1 Predator–Prey Robotics

Predator–prey systems have become a prominent research domain within evolutionary robotics because they naturally combine pursuit, evasion, adaptation, and cooperation. In such systems, predators must pursue evasive prey while continuously adapting to changes in both the environment and the behaviour of other agents. When multiple predators are involved, successful capture often requires coordination and cooperation among team members, particularly when the prey possesses superior speed or manoeuvrability. These characteristics make predator–prey tasks a valuable testbed for studying the emergence of complex behaviours in autonomous agents.

One of the earliest studies in evolutionary predator–prey robotics was conducted by Nolfi and Floreano (1998), who investigated the co-evolution of predator and prey robots. Their work demonstrated that increasingly sophisticated behaviours can emerge through competitive evolutionary processes, leading to an evolutionary "arms race" between predator and prey agents. Rather than being explicitly programmed, pursuit and evasion strategies evolved through repeated interactions.

Floreano et al. (1998) extended previous work on competitive co-evolution by investigating its application in real-world robotic systems. Their results demonstrated that competitive co-evolution can generate effective predator and prey behaviours and provided practical evidence for the feasibility of evolutionary robotics approaches.

More than a decade later, Nolfi (2012) revisited predator–prey robotics and reviewed the insights gained from earlier co-evolutionary studies. He argued that predator–prey systems remain valuable experimental platforms for investigating adaptation, competition, and the emergence of complex behaviours through evolutionary processes. This work reinforced the significance of predator–prey scenarios as a benchmark for evolutionary robotics research.

Among more recent contributions, Lan et al. (2019) presented one of the studies most closely related to this thesis. Their work investigated predator robots controlled by evolved neural networks in both simulation and physical robotic environments using Thymio robots. Through a two-stage evolutionary process, they demonstrated that effective predator pursuit strategies can be evolved in simulation and successfully transferred to physical robots. Similarly, Chen and Gao (2020) proposed a

framework for learning predator–prey behaviour across simulation and real-world environments using evolutionary methods integrated with ROS and Gazebo. These studies demonstrate that predator–prey robotics remains an active research area and provide methodological foundations for the present work.

A common characteristic of predator–prey robotics studies is that predator controllers rely on sensory information to perceive the environment and other agents. Some approaches assume broad access to global state information. For example, Sun et al. (2020) make the positions of all predator and prey robots visible to the other robots. Although this can simplify controller evolution, such information is rarely available in real-world robotic systems, which must instead rely on onboard sensors that provide only partial observations of the environment. Understanding how different observation modalities influence the emergence of predator behaviour therefore remains an important research question and motivates the present study.

2.2 Cooperation in Multi-Agent Systems

A central challenge in predator–prey robotics is cooperation between multiple predators. Capturing a moving prey often requires coordinated behaviour rather than independent pursuit, particularly when the prey can adapt its movements in response to the predators.

Several studies have investigated how cooperation emerges through evolutionary processes. Yong and Miikkulainen (2001) demonstrated that cooperative co-evolution can lead to the emergence of complementary behavioural roles within predator teams, allowing agents to develop specialized behaviours without explicit role assignment. In their prey-capture experiments, the co-evolved controllers were more efficient and robust than a single centralized controller, and effective cooperation emerged without explicit communication.

Building upon findings from studies such as these, Nitschke (2005) reviewed research on emergent cooperation in biologically inspired multi-agent systems, including predator–prey domains. He argued that coordinated behaviour can arise through evolutionary and self-organizing processes rather than explicit centralized control, highlighting the potential of biologically inspired approaches for generating cooperative behaviour in complex systems.

Communication has also been investigated as a mechanism for improving agent coordination. Jim and Giles (2000) evolved communication languages for predator agents and demonstrated that communication can significantly improve team performance in predator–prey tasks. Their results showed that allowing agents to exchange information helps overcome limitations caused by partial observability and facilitates more effective cooperation.

Related evidence for the emergence of coordinated behaviour was provided by Nelson et al. (2004), who evolved neural-network controllers for teams of mobile robots competing in a "Capture the Flag" task. Their work demonstrated that evolutionary methods can generate effective team behaviours in competitive multi-agent environments without requiring manually designed coordination strategies.

More recently, Sun et al. (2020) proposed a cooperative co-evolutionary approach for multi-robot pursuit in which predator agents were assisted by virtual agents that explored candidate actions in their local neighbourhoods. Their results demonstrated that cooperative evolutionary optimization can generate effective coordination strategies for predator teams without requiring explicit communication or manually designed pursuit behaviours. The study further highlighted the scalability of evolutionary approaches for coordinating larger groups of predator robots.

Collectively, these studies suggest that successful predator behaviour depends not only on the capabilities of individual agents but also on their ability to coordinate effectively with teammates. Furthermore, the literature demonstrates that such coordination can emerge through evolutionary processes, with or without explicit communication mechanisms. This observation motivates the use of multiple predators in the present work and highlights the importance of understanding how different observation methods influence the emergence of cooperative pursuit behaviour.

2.3 Simulation and the Reality Gap

Simulation is widely used in evolutionary robotics because it allows thousands of controller evaluations to be performed efficiently. However, an important challenge is the reality gap, referring to differences between behaviour observed in simulation and behaviour observed on physical robots.

Jakobi et al. (1995) were among the first researchers to identify this issue. They demonstrated that controllers evolved in simulation can exploit inaccuracies in the simulator and therefore perform poorly when transferred to real hardware. Their work highlighted a fundamental limitation of simulation-based optimization and became one of the foundational studies in sim-to-real robotics.

Building upon this idea, Koos et al. (2013) proposed the transferability approach, which explicitly considers differences between simulation and reality during optimization. Their work demonstrated

that incorporating transferability information can improve the deployment of evolved controllers on physical robots.

The importance of the reality gap is also reflected in more recent predator–prey studies. Both Lan et al. (2019) and Chen and Gao (2020) investigated approaches that combine simulation and physical robotics, demonstrating the continued relevance of sim-to-real transfer in evolutionary robotics.

Although the present thesis is conducted entirely in simulation, the reality-gap literature remains relevant because it highlights the importance of realistic sensing assumptions. Observation models based on exact state information may produce strong results in simulation while relying on information that would not normally be available to physical robots.

2.4 Research Gap

The literature reviewed in this thesis shows that evolutionary algorithms can generate effective predator behaviours, that coordination can emerge in multi-agent pursuit tasks, and that simulation can support the optimization of robot controllers (Floreano et al., 1998; Jim and Giles, 2000; Lan et al., 2019; Nolfi and Floreano, 1998; Sun et al., 2020; Yong and Miikkulainen, 2001). However, this literature primarily focuses on controller evolution, cooperation, communication, competitive co-evolution, or sim-to-real transfer. Within the reviewed work, the observation model is generally treated as part of the experimental setup rather than as the primary variable of comparison. Consequently, direct controlled comparisons between privileged information, panoramic vision, and restricted first-person vision appear to be limited.

To address this gap, the present thesis compares three observation models: privileged observation, 360° camera observation, and first-person view (FPV). Using an otherwise shared experimental setup, the study provides an exploratory comparison of the controller performance observed under these three representations.

3 Experimental Setup

This section describes the simulation environment, robot controllers, observation models, fitness function, evolutionary procedure, and evaluation protocol. Three experimental conditions are compared: privileged observation, first-person view, and 360-degree vision. All other components and parameter settings are held constant so that the observation model remains the primary experimental difference between the three conditions.

3.1 Simulation Environment

The experiments are conducted in Gazebo Classic using simulated Thymio II differential-drive robots. The environment consists of a $2\text{ m} \times 2\text{ m}$ square arena enclosed by walls and containing no internal obstacles. Three red robots act as predators and one green robot acts as the prey.

Every episode begins from the same initial configuration. The prey is positioned at the centre of the arena, while the predators are placed parallel to one another near the southern wall at approximately

$$(-0.55, -0.75), \quad (0, -0.75), \quad (0.55, -0.75).$$

All coordinates are expressed in metres. All robots initially face the positive y -direction, towards the prey. Fixed starting positions and orientations are used deliberately to isolate the effect of the observation model and to avoid advantages or disadvantages caused by randomly generated starting configurations.

Before fitness measurement begins, a startup period of 2 s of simulation time allows the controller processes and ROS 2 communication channels to initialize. Each evaluation episode subsequently lasts 30 s. The same environment, robot configuration, starting positions, and episode duration are used for all observation models.

Predator movement is limited to a maximum forward speed of 0.08 m/s and a maximum angular speed of 1.0 rad/s. The corresponding prey limits are 0.12 m/s and 1.0 rad/s, respectively.

The experiments are executed on Ubuntu 22.04 LTS using ROS 2 Humble and Gazebo Classic 11.10.2. The computer contains an Intel Core i7-10750H processor, 16 GB of DDR4 memory, and an NVIDIA GeForce RTX 2070 Mobile GPU.

3.2 Predator Controller

The predator controller is a feedforward neural network based on the compact controller used by Lan et al. (2019). It contains three input neurons, one hidden layer with four neurons, and two output neurons. Including all weights and biases, the controller contains

$$(3 \times 4) + 4 + (4 \times 2) + 2 = 26 \quad (1)$$

trainable parameters.

All three predators use the same network parameters during an evaluation, resulting in a homogeneous predator team. No individual predator is assigned a predefined role. Any coordinated behaviour must therefore emerge from the interaction between the shared controller, the observations received by each robot, and the environment.

For an input vector x , the hidden-layer activations are

$$h = \tanh(xW_1 + b_1), \quad (2)$$

where W_1 and b_1 denote the hidden-layer weights and biases. The network output is

$$o = \tanh(0.25(hW_2 + b_2)), \quad (3)$$

where W_2 and b_2 denote the output-layer weights and biases.

The factor 0.25 was introduced after preliminary experiments showed that large evolved weights frequently caused both output neurons to saturate. This produced nearly identical wheel commands and persistent straight-line movement. Scaling the output-layer pre-activations reduced this tendency while preserving the bounded output range of the hyperbolic tangent function.

The two outputs determine the angular velocities of the left and right wheels. These wheel commands are converted into linear and angular robot velocities using differential-drive kinematics.

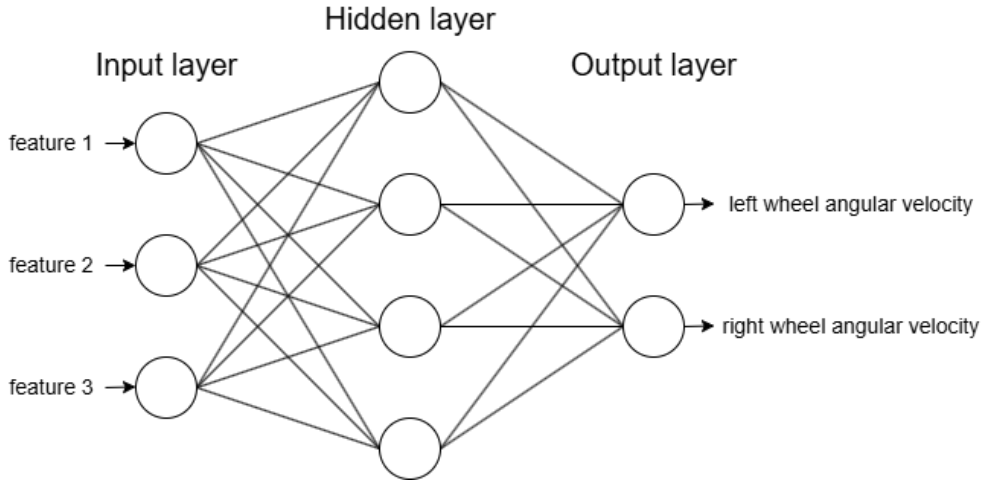


Fig. 1: Feedforward neural-network architecture used by the predator controllers. The three input features depend on the observation model, while the two outputs determine the angular velocities of the left and right wheels. Bias nodes are omitted for clarity.

3.3 Observation Models

The three observation models each provide three values to the neural network. Consequently, the network architecture and number of optimized parameters remain identical across the experimental conditions. The models differ only in the source and representation of the available information.

Privileged Observation The privileged model receives geometric information calculated directly from the simulated robot poses. Its input vector is

$$x_{\text{priv}} = [r_s/2.828 \ \Delta\theta/\pi \ d/2.828], \quad (4)$$

where r_s is the signed distance to the nearest predator, $\Delta\theta$ is the relative bearing from the observing predator to the prey, and d is the predator–prey distance. A positive value of r_s indicates that the nearest teammate is on the left of the observing predator, while a negative value indicates that it is on the right. Distances are normalized by the approximate arena diagonal, and the prey bearing is normalized to the interval $[-1, 1]$. This model therefore receives direct, noise-free geometric information.

First-Person View The first-person view condition uses only the front-facing camera of each predator. Green image regions represent the prey, while red regions represent neighbouring predators. Colour segmentation is used to identify the largest visible green and red blobs.

For each detected blob, x_b denotes its normalized horizontal image position and A_b its area normalized by the total image area. The visual input vector is

$$x_{\text{FPV}} = [-x_r\sqrt{A_r} - x_p\sqrt{A_p}], \quad (5)$$

where the subscripts r and p refer to the red teammate blob and green prey blob, respectively. The first value combines the side and apparent size of the largest visible teammate. The second represents the horizontal direction of the prey, and the third provides a visual closeness measure based on prey blob size. When the prey or a teammate is not visible, the corresponding values are set to zero. The limited field of view therefore makes this condition partially observable.

360-Degree Vision The 360-degree condition uses four cameras facing forward, backward, left, and right. Their images are stitched horizontally to form a panoramic representation. The rear image is divided between the two edges of the panorama, resulting in the following left-to-right order:

rear-right half | left | front | right | rear-left half

Placing the front view at the centre makes the centre of the panorama correspond approximately to the forward direction of the robot. The same colour-segmentation procedure and visual feature vector defined in Equation 5 are applied to the combined image. The 360-degree model retains the limitations of camera-based perception but substantially reduces the blind region present in the first-person view condition.

3.4 Prey Controller

The prey is controlled by a fixed, non-evolved Gaussian danger-field controller based on the strategy proposed by Lan et al. (2019). Keeping the prey controller fixed ensures that every predator controller and observation model is evaluated against the same opponent.

The controller constructs a danger field from the positions of the predators and the arena walls. Predator danger is represented using two-dimensional Gaussian functions centred on the predator positions, while one-dimensional Gaussian functions represent danger near the walls. The relative influence of predator and wall avoidance is controlled by the parameter α .

At each control step, the prey calculates the gradient of the combined danger field at its current position and moves in the opposite direction, corresponding to a locally less dangerous region. Its angular velocity depends on the difference between its current orientation and the desired escape direction. Its forward velocity is larger when it is aligned with that direction and smaller while turning.

The parameters σ_w and σ_p control the spatial spread of the Gaussian danger functions associated with the walls and predators, respectively. The parameter α controls the relative influence of predator danger compared with wall danger. Following Lan et al. (2019), the parameters are set to $\sigma_w = 0.2$, $\sigma_p = 0.25$, and $\alpha = 0.1$.

3.5 Fitness Function

The fitness function is adapted from the work of Lan et al. (2019). It rewards predators for remaining close to the prey while also discouraging the predator team from clustering.

Let $d_i(t_k)$ denote the Euclidean distance between predator i and the prey at sampling time t_k , and let $r_i(t_k)$ denote the distance between predator i and its nearest teammate. Fitness measurements are collected every 0.2s.

For a team containing N predators, the instantaneous fitness at sampling time t_k is defined as

$$f(t_k) = \frac{1}{N} \sum_{i=1}^N \frac{1}{\max(\varepsilon, d_i(t_k))} + \frac{1}{N} \sum_{i=1}^N r_i(t_k), \quad (6)$$

where $\varepsilon = 0.03$ m is included as a safeguard against division by zero. If an episode contains K valid samples, the final episode fitness is

$$F = \frac{1}{K} \sum_{k=1}^K f(t_k). \quad (7)$$

The first term increases when predators remain close to the prey, while the second term rewards separation between predators. Consequently, the fitness encourages predators to pursue the prey while occupying different positions rather than clustering together.

The original formulation of Lan et al. (2019) first averages the predator–prey distance over the complete episode and after that applies the reciprocal. This formulation was tested during preliminary experiments. Under the present experimental configuration, however, the spacing component provided an easily exploitable source of reward. Evolved controllers frequently increased their fitness by moving away from one another, often towards different arena boundaries, without consistently pursuing the prey.

For this reason, the reciprocal predator–prey distance is calculated at every sampling instant before the values are averaged over the episode. This modification increases the contribution of periods in which predators are close to the prey and produced behaviour that was more closely aligned with the intended prey pursuit task. The same adapted fitness function was fixed before the final comparative experiments and was applied identically to all three observation models.

3.6 Evolutionary Optimization

Inspired by Lan et al. (2019), the 26 neural-network parameters are optimized using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) (Hansen and Ostermeier, 2001). Each candidate solution is represented by a real-valued vector containing all network weights and biases.

The initial CMA-ES search distribution is centred on a zero vector with an initial step size of $\sigma = 0.5$. In each generation, 13 candidate controllers are sampled. Each candidate is evaluated in a single simulation episode, and the resulting episode fitness is used as its training fitness. One evaluation per candidate is used to keep the computational cost of training manageable.

Although all candidates are evaluated from the same initial robot configuration, individual fitness measurements may still be affected by nondeterminism in Gazebo physics, ROS 2 message timing, and process scheduling. This variability is not averaged out during training because each candidate is evaluated only once. However, the same evaluation procedure is applied consistently to all three observation models.

After all candidates in a generation have been evaluated, CMA-ES updates the mean, covariance matrix, and step size of its search distribution. Since the employed implementation minimizes its objective, the negative candidate fitness is supplied to the optimizer as the loss.

Training is performed for 100 generations, resulting in 1300 candidate evaluations and 1300 simulation episodes per observation model. One complete CMA-ES run is conducted for each of the three observation models. Throughout each run, the five controllers with the highest training fitness encountered across all generations are retained for post-training evaluation.

A complete summary of the experimental parameters is provided in Table 2 in Appendix A.

3.7 Controller Selection and Evaluation

After training, each of the five retained controllers for an observation model is evaluated over ten selection episodes using the same fixed initial configuration and experimental parameters. The mean fitness of each controller is calculated, and the controller with the highest mean fitness is selected as the representative controller for that observation model.

The selected controller is then evaluated in an additional, separate set of ten episodes. These final evaluation episodes are not used during controller selection. The mean fitness and standard deviation obtained from this second set are reported as the final performance measures. The mean fitness represents the controller’s average performance, while the standard deviation indicates how consistently it performs under nondeterminism arising from Gazebo physics, ROS 2 message timing, and process scheduling.

The same selection and evaluation procedure is applied to the privileged, first-person view, and 360° camera observation models. All episodes use the same fixed starting positions, robot orientations, prey behaviour, and simulation parameters.

Only one evolutionary training run is conducted for each observation model. Consequently, the experiment compares the representative controllers produced by these three runs and does not estimate variability between multiple CMA-ES training runs. The repeated final evaluations measure execution-level variability of the selected controllers rather than variation in the evolutionary optimization process.

3.8 Statistical Analysis

The statistical analysis was performed on the fitness scores obtained from the ten final evaluation episodes of each selected controller. These evaluations produced three groups of fitness values corresponding to the privileged, 360° camera, and FPV observation models.

Because each group contained only ten observations and normality could not be assumed, the controllers were compared using the non-parametric Kruskal–Wallis test. The null hypothesis was that the fitness-score distributions of the three selected controllers did not differ. The significance level was set to $\alpha = 0.05$.

The magnitude of the overall difference was additionally described using the Kruskal–Wallis epsilon-squared effect size.

The statistical analysis concerns repeated evaluations of the three selected controllers. Since only one evolutionary training run was performed for each observation model, the test does not measure variability between multiple evolutionary runs and cannot establish the expected performance of each observation model across repeated training runs.

4 Experimental Results

This section presents the outcomes of evolutionary training, controller selection, and final evaluation for the privileged, 360° camera, and FPV observation models. One evolutionary run was conducted for each observation model. Consequently, the training curves describe the three observed optimization runs rather than average behaviour across multiple evolutionary runs.

4.1 Evolutionary Training

Figure 2 shows the best fitness discovered during each evolutionary run as a function of the cumulative number of fitness evaluations. Since every candidate was evaluated in exactly one simulation episode, the horizontal axis also represents the number of completed training episodes.

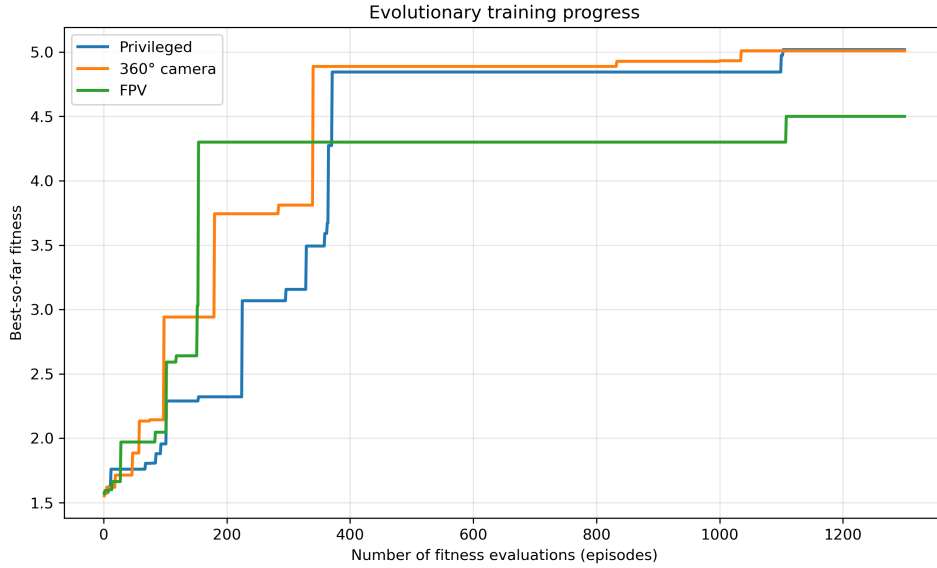


Fig. 2: Best-so-far fitness during evolutionary training. Each curve shows the highest fitness encountered up to the corresponding candidate evaluation during one CMA-ES run. Every candidate controller was evaluated in one simulation episode.

All three runs discovered controllers with substantially higher fitness than those evaluated at the beginning of training. The initial best fitness was approximately 1.6 for all three observation models. The privileged and 360° camera runs eventually reached best-so-far fitness values of approximately 5.0, while the FPV run reached approximately 4.5.

The improvements occurred mainly during the first part of training. The 360° camera run reached a fitness close to 4.9 after approximately 340 evaluations, while the privileged run reached a similar value after approximately 370 evaluations. The FPV run reached approximately 4.3 after around 150 evaluations, remained at this value for most of the run, and increased to approximately 4.5 during the final generations.

The horizontal plateaus in Figure 2 represent periods during which no newly evaluated candidate exceeded the highest fitness previously observed. Nevertheless, all three curves increased relative to

their initial values, showing that the evolutionary optimization discovered progressively higher-fitness controllers in each of the observed runs.

Figure 3 shows the population-level development over the 100 CMA-ES generations. The solid line in each panel represents the mean fitness of the 13 candidate controllers evaluated in that generation. The shaded region represents one standard deviation across the candidate fitness values, and the dashed line shows the best candidate in the generation.

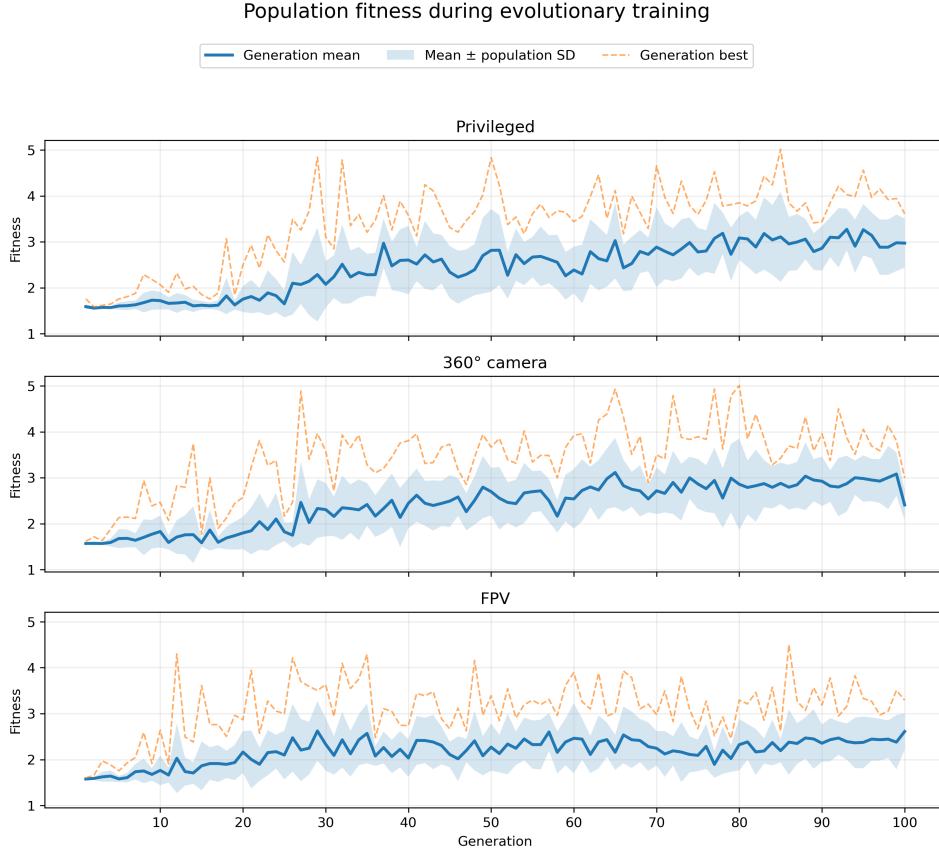


Fig. 3: Population fitness over 100 CMA-ES generations. Solid lines show the mean candidate fitness in each generation, shaded regions represent one standard deviation across candidates within the generation, and dashed lines show the best-performing candidate in each generation. Each panel represents one evolutionary run.

The mean population fitness increased relative to the initial generations for all three observation models. The privileged and 360° camera populations reached generation means close to 3.0 during the later stages of training. The FPV population improved during the earlier generations but generally remained between approximately 2.0 and 2.5 during the later part of the run.

The generation-best values fluctuated more strongly than the population means. Individual generations therefore sometimes contained one high-performing candidate while the remaining candidates achieved lower fitness. This also explains why the best-so-far values in Figure 2 were considerably higher than the generation means in Figure 3.

4.2 Controller Selection

After training, the five controllers with the highest recorded training fitness were retained for each observation model. Each retained controller was then evaluated over ten selection episodes, and the controller with the highest mean selection fitness was chosen for final evaluation.

For the privileged observation model, Controller 2 obtained the highest selection mean fitness of 3.16. For the 360° camera model, Controller 3 obtained the highest mean of 3.12, while for the FPV model, Controller 3 obtained the highest mean of 2.77.

The mean selection fitness was lower than the recorded single-episode training fitness for all three selected controllers. The privileged controller decreased from 4.97 to 3.16, the 360° camera controller from 4.92 to 3.12, and the FPV controller from 4.28 to 2.77. This reduction is consistent with the training fitness being based on one episode, whereas the selection fitness was averaged over ten repeated evaluations. The comparison therefore illustrates the importance of re-evaluating high-scoring candidate solutions.

Table 1: Controllers selected after ten selection episodes per retained controller.

Observation model	Selected controller	Recorded training fitness	Mean selection fitness
Privileged	2	4.97	3.16
360° camera	3	4.92	3.12
FPV	3	4.28	2.77

The complete results for all five retained controllers are presented in Figures 6–8 in Appendix B. The standard deviation intervals overlapped for several retained controllers. The selected controllers are therefore those with the highest observed mean under the predefined selection procedure, rather than controllers demonstrated to be statistically superior to all other retained candidates.

4.3 Final Controller Performance

Each selected controller was evaluated in a separate final set of ten episodes. These episodes were not used during controller selection. Figure 4 presents the resulting fitness values. The bars show the mean fitness, the error bars show one sample standard deviation, and the points show the ten individual episode scores.

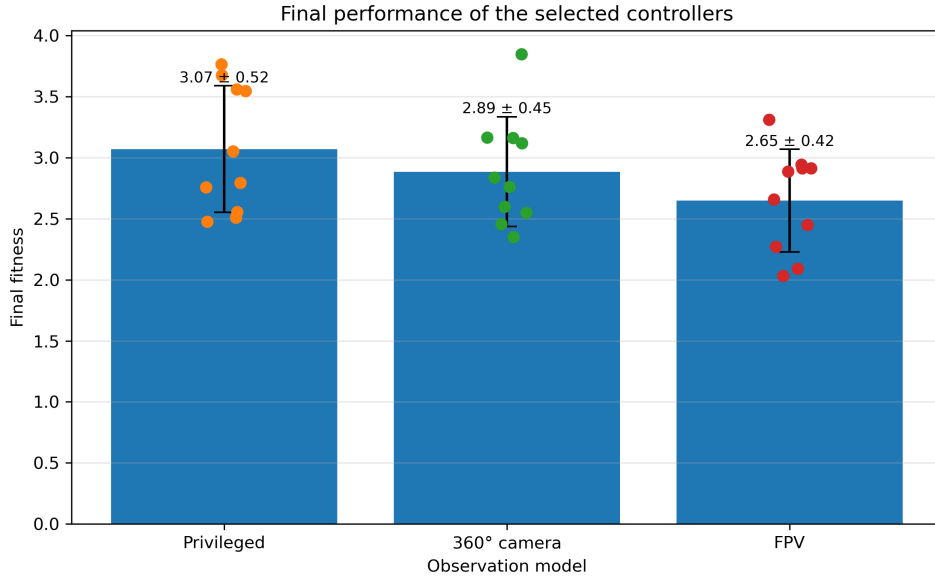


Fig. 4: Final performance of the selected controllers over ten evaluation episodes. Bars show mean fitness, error bars show one sample standard deviation, and points show the individual episode scores.

The privileged controller achieved the highest observed mean final fitness, with 3.07 ± 0.52 . The 360° camera controller achieved 2.89 ± 0.45 , while the FPV controller achieved 2.65 ± 0.42 . The values following the means represent the sample standard deviation across the ten final evaluation episodes.

Based on the observed mean fitness, the controller ordering was

$$F_{\text{privileged}} > F_{360^\circ} > F_{\text{FPV}}.$$

The difference between the privileged and 360° camera means was 0.18. The difference between the 360° camera and FPV means was 0.24, while the difference between the privileged and FPV means was 0.42. However, Figure 4 also shows substantial overlap between the individual episode scores of the three controllers.

Representative videos of the final selected controllers are available for all three observation models. The three example videos illustrate the behaviour of the homogeneous predator team controlled by the selected privileged, 360° camera, and FPV controllers, respectively. The videos were recorded after the evaluation procedure and are included only as qualitative demonstrations of the evolved behaviours.

- Privileged observation model
- 360° camera observation model
- First-person view (FPV) observation model

4.4 Statistical Comparison

A Kruskal–Wallis test was applied to the final fitness scores of the three selected controllers. The test did not detect a statistically significant difference between the three fitness-score distributions,

$$H(2) = 2.511, \quad p = 0.285.$$

The estimated Kruskal–Wallis epsilon-squared effect size was

$$\varepsilon^2 = 0.019.$$

The null hypothesis was therefore not rejected at the predefined significance level of $\alpha = 0.05$. Although the privileged controller had the highest observed mean and the FPV controller had the lowest, the final evaluation did not provide sufficient statistical evidence that the distributions differed.

This test compares repeated execution of the three selected controllers. It does not measure variation between evolutionary runs because only one CMA-ES run was conducted for each observation model.

4.5 Representative Trajectories

Figure 5 shows one representative final evaluation episode for each selected controller. For each model, the displayed episode was selected as the episode whose fitness was closest to the median of the ten final evaluation scores. This avoids selecting only the highest-performing or most visually favourable episode.

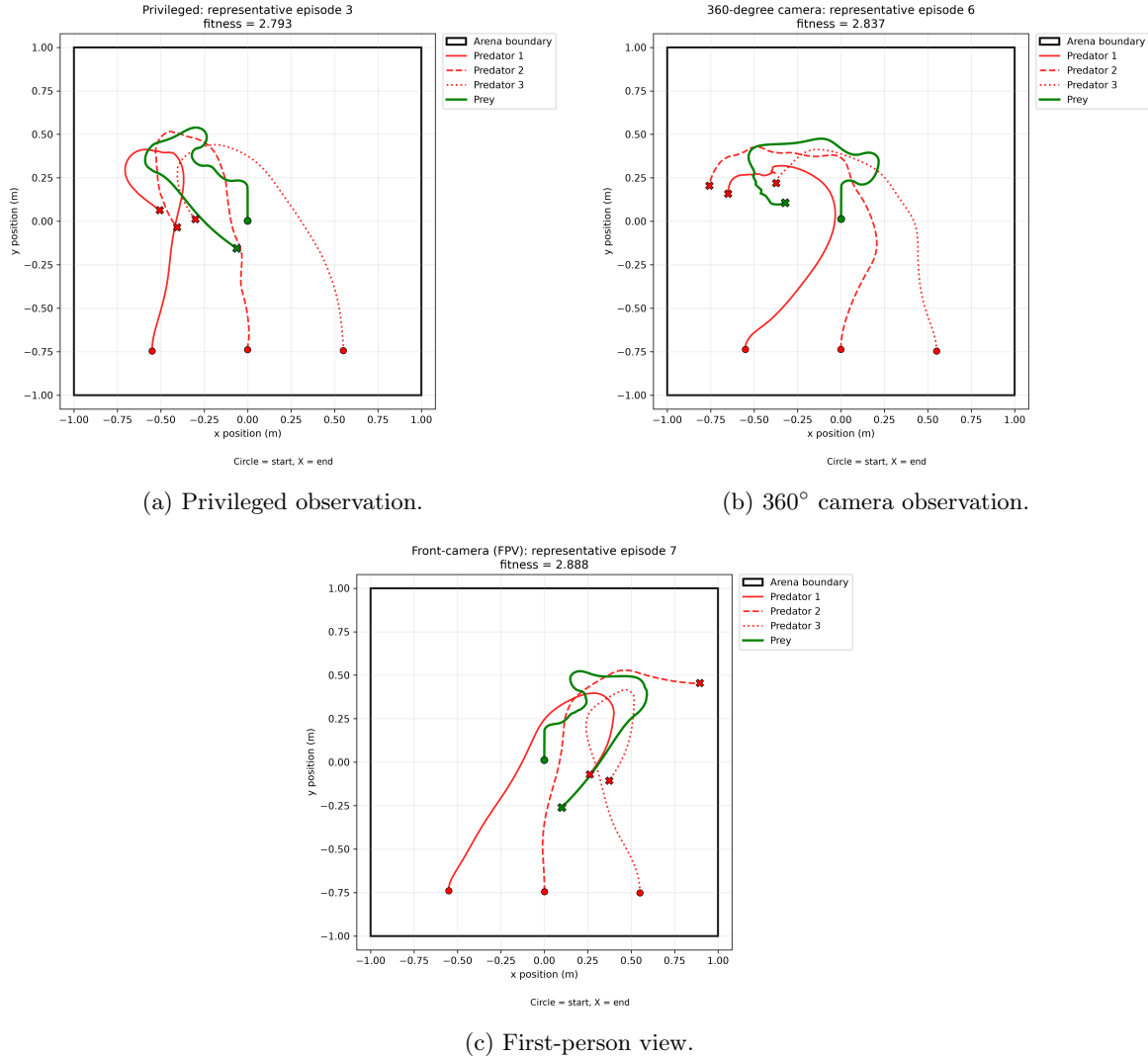


Fig. 5: Representative median-fitness trajectories. Red lines show predator trajectories and the green line shows the prey trajectory.

In the representative privileged episode, all three predators moved from their initial positions towards the prey's trajectory and finished within the same general region. The representative 360°

camera episode showed a similar pattern, although the predators’ final positions were more dispersed around the prey’s path.

In the representative FPV episode, the predators also moved towards the region occupied by the prey. Two predators finished relatively close to the prey’s final trajectory, while the third continued farther into the upper-right region of the arena. These plots provide qualitative examples of the behaviours produced by the selected controllers. Because only one episode is shown for each controller, they are treated as illustrative examples rather than evidence of systematic behavioural differences between the observation models.

5 Analysis and Discussion

This study investigated how the observation representation available to a homogeneous predator team influenced the evolution and performance of its controllers. Under the tested conditions, CMA-ES produced increasingly high-fitness controllers for the privileged, 360° camera, and first-person view observation models. The selected privileged controller achieved the highest observed mean final fitness, followed by the 360° camera controller and the FPV controller.

However, the statistical comparison did not detect a significant difference between the final fitness-score distributions. The results should therefore be interpreted as an exploratory comparison between three controllers produced by three distinct evolutionary runs. They do not establish that one observation model generally produces superior controllers.

5.1 Influence of the Observation Model

The selected privileged controller achieved the highest observed mean final fitness of 3.07. A possible explanation is that its inputs provided direct and noise-free information about the prey and the nearest teammate. The prey distance and bearing remained available regardless of the predator’s orientation, and the controller did not need to infer these properties from camera output. This resulted in a comparatively stable and informative representation of the current situation.

The selected 360° camera controller achieved a slightly lower mean fitness of 2.89. Its observed performance was therefore close to that of the privileged controller despite relying on visual observations rather than direct simulator state information. The panoramic image substantially reduced the blind region present in the FPV condition, allowing predators to detect the prey and neighbouring robots from most directions.

Nevertheless, the 360° camera representation remained less precise than the privileged representation. The prey distance was represented by its apparent blob size rather than an exact geometric distance, while its direction was estimated from its horizontal position in the stitched image. The resulting features also depended on colour segmentation and on the construction of the panoramic image. These differences may partly explain why the observed mean fitness was lower than that of the privileged controller. However, the difference between their final means was only 0.18, and their individual episode scores overlapped considerably.

The selected FPV controller achieved the lowest observed mean final fitness of 2.65. In this condition, the prey and neighbouring predators could only be detected when they were within the field of view of the front-facing camera. When the prey was not visible, the corresponding input values were set to zero. The same value could therefore represent different situations, such as the prey being behind, to the left, or to the right of the robot.

This ambiguity made the FPV condition more strongly affected by partial observability than the other observation conditions. After turning away from the prey, a predator received no direct information about the direction in which it should turn to regain visual contact. The employed neural network contained no memory of previous observations. It could therefore not explicitly retain the last observed prey direction after the prey left the camera image.

The representative FPV trajectory is consistent with this possible difficulty. Two predators finished relatively close to the prey’s trajectory, while the third continued farther into another region of the arena. However, the trajectory alone does not establish whether this behaviour was caused by loss of visual contact.

5.2 Evolutionary Training and Controller Selection

The best-so-far fitness increased substantially during all three evolutionary runs. The privileged and 360° camera runs reached individual training fitness values close to 5.0, while the FPV run reached approximately 4.5. The generation-level population means also increased relative to the initial generations, indicating that the improvement was not restricted to one isolated candidate.

At the same time, the generation-best fitness values fluctuated considerably more than the population means. Individual candidates occasionally achieved a high fitness even when the remaining population performed substantially worse. This is important because each candidate was evaluated in only one episode during training. A single fitness measurement could therefore be influenced by execution-level nondeterminism in Gazebo physics, ROS 2 message timing, and process scheduling.

The difference between the recorded training fitness and the subsequent selection fitness further demonstrates this issue. The privileged controller decreased from a single-episode training fitness of 4.97 to a ten-episode selection mean of 3.16. The 360° camera controller decreased from 4.92 to 3.12, while the FPV controller decreased from 4.28 to 2.77.

These reductions show that the original single-episode values were optimistic estimates of repeated controller performance. Retesting the five retained controllers over ten episodes was therefore important because it reduced the influence of unusually favourable individual training episodes.

5.3 Statistical and Behavioural Interpretation

The observed final mean fitness values followed the order privileged, 360° camera, and FPV. However, the Kruskal–Wallis test produced

$$H(2) = 2.511, \quad p = 0.285,$$

and the estimated epsilon-squared effect size was

$$\epsilon^2 = 0.019.$$

The null hypothesis was therefore not rejected at the significance level $\alpha = 0.05$. The available final evaluation data did not provide sufficient evidence that the three fitness-score distributions differed.

This result does not demonstrate that the controllers were equivalent. Each group contained only ten episode scores, and the observations overlapped substantially. More importantly, the statistical test compared repeated executions of only three selected controllers. It did not compare controllers obtained from multiple evolutionary runs.

The main independent unit in a broader comparison of observation models should be the evolutionary run, not an individual execution episode of one selected controller. Repeating the same controller ten times measures the variability of that controller under simulation nondeterminism. It does not measure how consistently an observation model enables CMA-ES to produce a high-performing controller.

Therefore, this work cannot support broad statistical claims about which observation model can be considered generally better. It establishes that three distinct observation models were implemented and compared under the same controlled experimental setup. In the three observed runs, the privileged controller had the highest mean performance and the FPV controller had the lowest, but the experiment did not establish a statistically supported general ranking of the observation models.

The representative trajectories show that all three selected controllers produced pursuit-like movement. In each displayed episode, the predators moved from their initial positions towards the region occupied by the prey. The privileged and 360° camera examples showed the predators finishing in the same general region as the prey’s path. The FPV example also showed movement towards the prey, although the predators’ final positions were more dispersed.

These trajectory plots should be treated only as qualitative illustrations. Only one episode was shown for each controller, selected because its fitness was closest to the median final-evaluation score. They illustrate the types of behaviour produced by the selected controllers but do not provide independent evidence of systematic behavioural differences.

5.4 Simulation Platform Considerations

Gazebo Classic and ROS 2 provided a useful development environment for constructing the simulated robots, implementing the camera observations, and observing the interaction between the predators and the prey. Gazebo was initially selected partly because its graphical interface made the experiments visually appealing and made controller behaviour easier to inspect and debug.

However, in the present implementation, the combination of Gazebo Classic and ROS 2 was not an ideal platform for the large number of repeated evaluations required by evolutionary optimization. Each candidate evaluation required the simulator, controllers, sensors, and ROS 2 communication processes to be initialized and coordinated. Performing 1300 training episodes per observation model was therefore computationally and operationally expensive.

The setup also displayed execution-level nondeterminism. Repeated evaluations of the same controller from the same initial configuration could produce different fitness values because of differences

in physics execution, ROS 2 message timing, callback timing, and operating-system process scheduling. This variability complicated controller evaluation and increased the need for repeated episodes during the selection and final evaluation stages.

These observations do not establish that Gazebo and ROS 2 are unsuitable for evolutionary robotics in general. They indicate that, at least in the employed setup, their computational overhead and nondeterministic execution limited the number of evolutionary runs that could reasonably be performed. The graphical interface was useful during development and qualitative inspection, but a lighter simulation platform may be more appropriate for experiments requiring many thousands of evaluations.

5.5 Limitations

The most important limitation is that only one evolutionary run was conducted for each observation model. Evolutionary optimization is stochastic, and another CMA-ES run could produce a different controller and a different ordering of final performance. The number of runs was limited by the substantial computational and operational cost of the Gazebo–ROS 2 experimental setup. The repeated final episodes measure execution-level variability of the selected controllers, but they do not measure variability in the evolutionary process itself.

A second limitation is the use of fixed initial positions and orientations. This controlled configuration ensured that all candidates faced the same initial task and helped isolate the observation representation. However, it also allowed controllers to specialize to one starting arrangement. A more complex environment might reveal differences between the three observation models that were not apparent in the relatively simple setup used in this study.

Although all three observation models supplied three inputs to the same neural network architecture, the inputs represented different types of information. The privileged model directly encoded geometric quantities, including prey distance and bearing, whereas the camera-based models encoded features derived from blob position and apparent size. This difference was intentional because the study compared complete observation models. However, it means that any observed performance differences cannot be attributed to a single property, such as observation precision, visual coverage, or partial observability.

A further limitation concerns the fixed prey controller. The relatively low value of α gave wall avoidance a strong influence compared with predator avoidance. As a result, the prey sometimes appeared to prioritise remaining away from the arena boundaries even when moving closer to a wall could have provided a more effective escape route from the predators. This may have made some pursuit episodes easier for the predators.

Finally, the work was conducted entirely in simulation. The simulated cameras, robot dynamics, communication timing, and visual conditions do not fully represent physical Thymio robots. The results therefore do not demonstrate that the evolved controllers would transfer successfully to real hardware.

5.6 Future Work

The highest priority for future work is to conduct multiple evolutionary runs for every observation model. Each run should produce its own selected controller, which should then be evaluated on a separate set of episodes. Statistical testing could subsequently compare the results across multiple evolutionary runs rather than only repeated executions of one controller.

Alternative simulation platforms should also be investigated. A simulator such as ARGoS may provide a lighter environment for large-scale multi-robot evolutionary experiments. Its simulation speed, robot and sensor models, compatibility with the required observations, and reproducibility would need to be assessed before replacing the current Gazebo–ROS 2 implementation.

Recurrent neural networks or other controllers with memory could be evaluated, particularly for the FPV condition. Memory could allow a predator to retain the last observed prey direction after the prey leaves the camera image and could support more effective search behaviour.

Future work could also compare CMA-ES with an evolutionary algorithm using explicit elitist survivor selection. Preserving the highest-performing controllers between generations may produce different optimization dynamics and could reduce the risk of strong solutions being lost during the search.

The experiments should additionally be repeated in a larger arena. A larger environment would create a more realistic pursuit scenario and reduce the relative influence of the arena walls on predator and prey behaviour. This would be particularly relevant for evaluating whether the observed behaviours generalize beyond the compact $2\text{ m} \times 2\text{ m}$ setting.

Finally, the selected controllers could be tested on physical Thymio robots. Real-world experiments would reveal how robust the visual observations and evolved behaviours are to lighting variation, camera noise, communication delays, wheel differences, and other aspects of the reality gap.

6 Conclusion

This thesis compared the performance of evolved predator controllers under three observation representations in a simulated predator–prey task. Three observation models were compared: privileged geometric information, 360° camera observations, and first-person view (FPV). The same simulation environment, neural-network architecture, fitness function, prey controller, and evolutionary procedure were used in all three conditions.

CMA-ES improved controller fitness during each of the three evolutionary runs, showing that effective pursuit-like behaviour could be evolved under all three observation models. In the final evaluation, the selected privileged controller achieved the highest observed mean fitness of 3.07, followed by the 360° camera controller with 2.89 and the FPV controller with 2.65. This ordering suggests that direct geometric information and broader visual coverage may have provided an advantage under the tested conditions, while the restricted field of view and partial observability of FPV may have made the task more difficult.

However, the Kruskal–Wallis test did not detect a statistically significant difference between the final fitness-score distributions. In addition, only one evolutionary run was conducted for each observation model. Consequently, the results do not establish a general ranking of the three observation models. Instead, they provide an exploratory comparison of the controllers produced by the three observed runs.

The observed differences suggest that the observation representation may influence the behaviour and performance of evolved predator controllers under the tested conditions, but stronger conclusions require multiple evolutionary runs and evaluation across more varied initial conditions and environments.

Code Availability

The source code used in this study is available at <https://github.com/Vseonym/Predator-Prey-Robotics>.

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Bibliography

- Jiunhan Chen and Zhenyu Gao. A framework for learning predator–prey agents from simulation to real world. *arXiv preprint arXiv:2010.15792*, 2020. <https://doi.org/10.48550/arXiv.2010.15792>.
- Dario Floreano, Stefano Nolfi, and Francesco Mondada. Competitive co-evolutionary robotics: From theory to practice. In Rolf Pfeifer, Bruce Blumberg, Jean-Arcady Meyer, and Stewart W. Wilson, editors, *From Animals to Animats 5: Proceedings of the Fifth International Conference on Simulation of Adaptive Behavior*, pages 515–524, Cambridge, MA, 1998. MIT Press.
- Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9(2):159–195, 2001. <https://doi.org/10.1162/106365601750190398>.
- Nick Jakobi, Phil Husbands, and Inman Harvey. Noise and the reality gap: The use of simulation in evolutionary robotics. In Federico Morán, Alvaro Moreno, Juan Julián Merelo, and Pablo Chacón, editors, *Advances in Artificial Life*, volume 929 of *Lecture Notes in Computer Science*, pages 704–720, Berlin, Heidelberg, 1995. Springer. https://doi.org/10.1007/3-540-59496-5_337.
- Kam-Chuen Jim and C. Lee Giles. Talking helps: Evolving communicating agents for the predator-prey pursuit problem. *Artificial Life*, 6(3):237–254, 2000. <https://doi.org/10.1162/106454600568861>.
- Sylvain Koos, Jean-Baptiste Mouret, and Stéphane Doncieux. The transferability approach: Crossing the reality gap in evolutionary robotics. *IEEE Transactions on Evolutionary Computation*, 17(1):122–145, 2013. <https://doi.org/10.1109/TEVC.2012.2185849>.
- Gongjin Lan, Jiunhan Chen, and A. E. Eiben. Simulated and real-world evolution of predator robots. In *2019 IEEE Symposium Series on Computational Intelligence (SSCI)*, pages 1974–1981. IEEE, 2019. ISBN 978-1-7281-2486-5. <https://doi.org/10.1109/SSCI44817.2019.9002696>.
- A. L. Nelson, E. Grant, and T. C. Henderson. Evolution of neural controllers for competitive game playing with teams of mobile robots. *Robotics and Autonomous Systems*, 46(3):135–150, 2004. <https://doi.org/10.1016/j.robot.2004.01.001>.
- Geoff Nitschke. Emergence of cooperation: State of the art. *Artificial Life*, 11(3):367–396, 2005. <https://doi.org/10.1162/1064546054407194>.
- Stefano Nolfi. Co-evolving predator and prey robots. *Adaptive Behavior*, 20(1):10–15, 2012. <https://doi.org/10.1177/1059712311426912>.
- Stefano Nolfi and Dario Floreano. Coevolving predator and prey robots: Do arms races arise in artificial evolution? *Artificial Life*, 4(4):311–335, 1998. <https://doi.org/10.1162/106454698568620>.
- Lijun Sun, Chao Lyu, and Yuhui Shi. Cooperative coevolution of real predator robots and virtual robots in the pursuit domain. *Applied Soft Computing*, 89:106098, 2020. <https://doi.org/10.1016/j.asoc.2020.106098>.
- Chern Han Yong and Risto Miikkulainen. Cooperative coevolution of multi-agent systems. Technical Report AI01-287, Department of Computer Sciences, University of Texas at Austin, 2001.

Appendix

A Experimental Parameters

Table 2 summarizes the principal parameters used throughout the experiments.

Table 2: Principal parameters used in the experiments.

Parameter	Value
Arena size	2 m $\times$ 2 m
Predators / prey	3/1
Episode duration	30 s
Startup period	2 s
Fitness sampling interval	0.2 s
Network architecture	3-4-2
Trainable parameters	26
Maximum predator speed	0.08 m/s
Maximum prey speed	0.12 m/s
Maximum angular speed	1.0 rad/s
CMA-ES population size	13
Initial CMA-ES step size	0.5
Generations	100
Evaluations per candidate	1
Training runs per model	1
Retained controllers	5
Selection episodes per retained controller	10
Final evaluation episodes per selected controller	10

B Additional Experimental Results

B.1 Retained-Controller Evaluation

The following figures show the ten-episode selection results for the five retained controllers from each observation model. Error bars represent one sample standard deviation, and hatched bars identify the controllers selected for final evaluation.

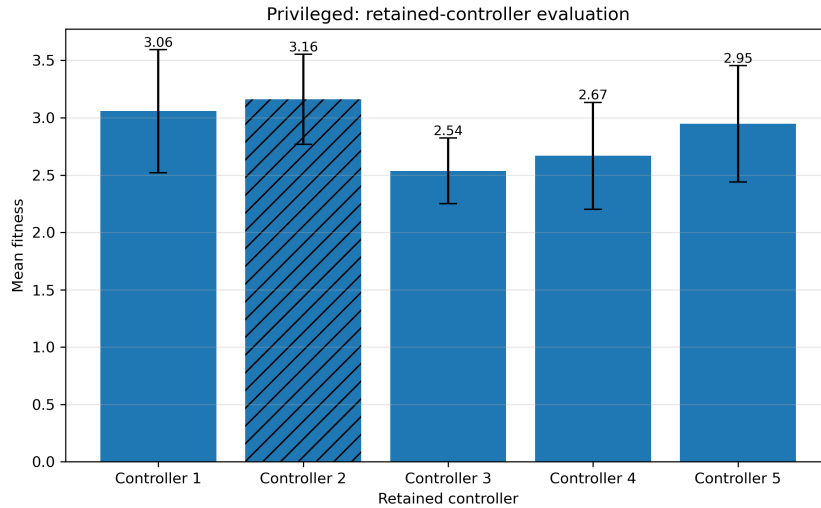


Fig. 6: Retained-controller evaluation for the privileged observation model. Controller 2 obtained the highest mean fitness and was selected for final evaluation.

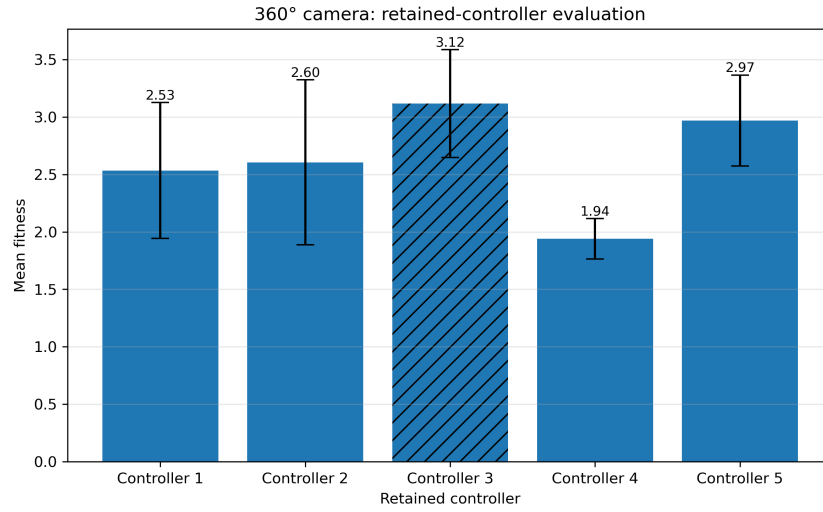


Fig. 7: Retained-controller evaluation for the 360° camera observation model. Controller 3 obtained the highest mean fitness and was selected for final evaluation.

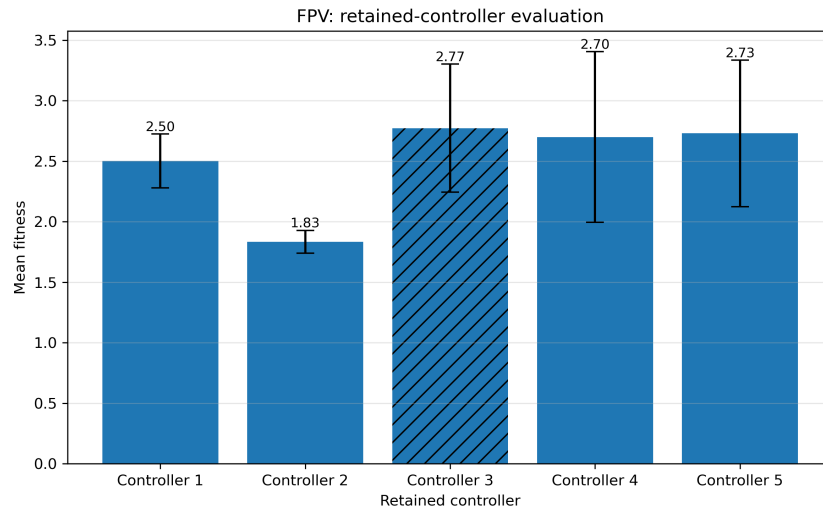


Fig. 8: Retained-controller evaluation for the FPV observation model. Controller 3 obtained the highest mean fitness and was selected for final evaluation.

C Picture of Robots in Gazebo

This section of the appendix shows pictures of predator and prey robots in Gazebo simulator.

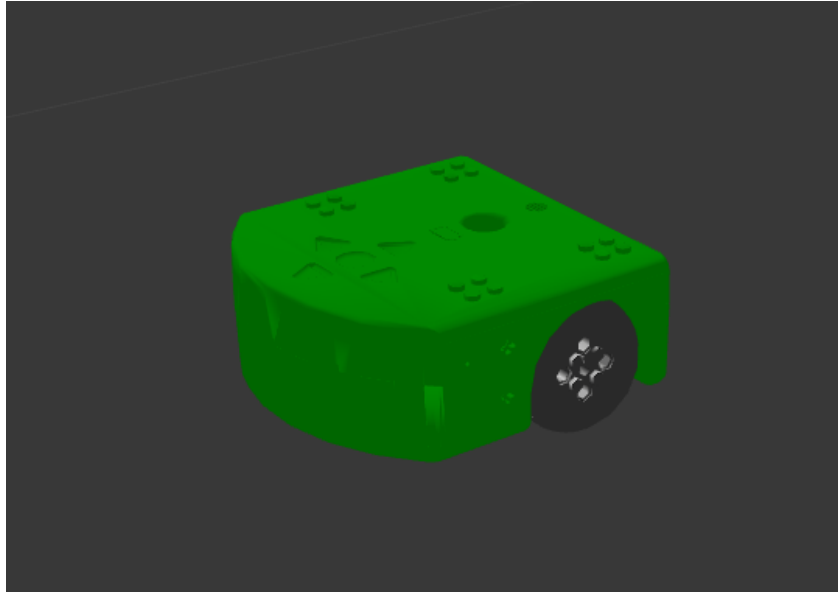


Fig. 9: Prey in Gazebo.

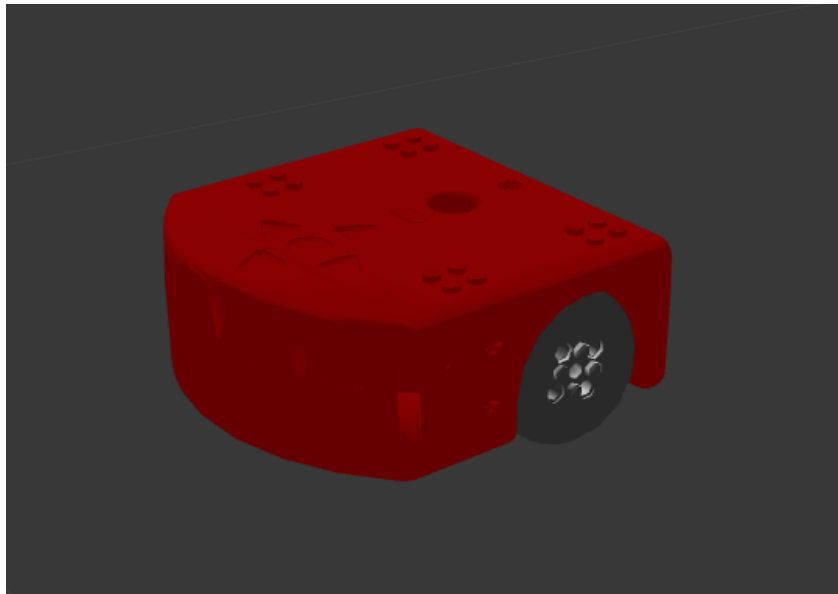


Fig. 10: Predator in Gazebo.